# COMP3438 System Programming

Course Logistics 2026 Fall Semester

# Instructor, Lecture Venue, and Time

438S01 Wed 8:30am-11:30am

Instructor: WANG Qixin

Lecture: PQ603, PQ604A/B/C

**Microsoft Teams meeting**

**Join:** https://teams.microsoft.com/meet/410971205034834?p=Nd7JY4Qvif0GqOqWka

Meeting ID: 410 971 205 034 834

Passcode: RL6cc35Q

# Instructor, Lecture Venue, and Time

438S02 Tue 12:30pm-3:30pm

Instructor: WANG Qixin (Week 1-9), TAYYAB Muhammad (Week 10-13)

Lecture: PQ603, PQ604A/C

**Microsoft Teams meeting**

**Join:** https://teams.microsoft.com/meet/491234356857378?p=aTDKHyhL9iordcn2dI

Meeting ID: 491 234 356 857 378

Passcode: Rn6nA9rA

# Instructor, Lecture Venue, and Time

438S03 Wed 12:30pm-3:30pm

Instructor: WANG Qixin (Week 1-9), TAYYAB Muhammad (Week 10-13)

Lecture: PQ603, PQ604B/C

**Microsoft Teams meeting**

**Join:** https://teams.microsoft.com/meet/490724421946861?p=72Afu9eeoSfhJ3URaF

Meeting ID: 490 724 421 946 861

Passcode: SZ2oi34m

# Office Hours

Instructor: WANG Qixin

S01 (Week 1-13): Wed 6:30pm-8:30pm

S02 (Week 1-9): Tue 6:30pm-8:30pm

S03 (Week 1-9): Tue 8:30pm-9:30pm, Wed 8:30pm-9:30pm

**Microsoft Teams meeting**

**Join:** https://teams.microsoft.com/meet/410971205034834?p=Nd7JY4Qvif0GqOqWka

Meeting ID: 410 971 205 034 834

Passcode: RL6cc35Q

# Contact Info

or Discussion Forum:

S01: Canvas -> COMP3438_2026271_A -> Discussions

S02 & S03: Canvas -> COMP3438_2026271_B -> Discussions

or (as the last resort) you can email me via csqwang@polyu.edu.hk with email title:

"Comp3438 [Your student ID] [Your @connect.polyu.hk preferred name] [optional content]"

Note I am not very responsive to emails, as I am overwhelmed by other duties. Also, emails without the above title may be discarded automatically by the junk email filters.

I need your @connect.polyu.hk "preferred name" (note it is NOT your connect account name) in order to find you in MS Teams.

# Teaching Assistants (random ordering)

PENG Xinghao, Email: xing-hao.peng@connect.polyu.hk
Preferred name: “PENG, Xing Hao [Student]”

WU Zekai, Email: zekai-candyore.wu@connect.polyu.hk
Preferred name: “WU, Zekai CandyOre [Student]”

ZHANG Hengjia, Email: heng-jia.zhang@connect.polyu.hk
Preferred name: “ZHANG, Heng Jia [Student]”

CHU Jinzhao, Email: jin-zhao.chu@connect.polyu.hk
Preferred name: “CHU, Jin Zhao [Student]”

SONG Haoyang, Email: haoyang.song@connect.polyu.hk
Preferred name: “SONG, haoyang [Student]”

HU Xu, Email: xu0529.hu@connect.polyu.hk
Preferred name: “HU, Xu0529 [Student]”

QIN Ziheng, Email: zi-heng.qin@connect.polyu.hk
Preferred name: “QIN, Zi Heng [Student]”

ZHANG Yibin, Email: yibin0914.zhang@connect.polyu.hk
Preferred name: “ZHANG, Yibin0914 [Student]”

LI Bairui, Email: bairui.li@connect.polyu.hk
Preferred name: “LI, bairui [Student]”

# Assessment

Continuous Assessment: 30%

Assignments (lab and/or other assignments): 30%

(Final) Examination (3 hours): 70%

# Late Assessment Policy

In case you apply for Late Assessment due to sickness or other reasons, below is the Late Assessment (i.e. makeup exam) Policy:

Problems and difficulty level of late assessment will surely be adjusted. Closed book exam, additional in-lab machine programming problems, and oral defence questions may be added. The guideline is that no one shall be allowed to profit on increased review time by taking late assessments.

# Labs

Session: S01

Venue: PQ603

Format: with TA face-to-face

| Lab ID | Lab Date and Hour | Week 2-7 | Week 8-13 |
|---|---|---|---|
| S01 Lab1 | Thu 4:30pm-5:20pm | CHU Jinzhao | CHU Jinzhao |
| S01 Lab2 | Thu 5:30pm-6:20pm | ZHANG Hengjia | ZHANG Hengjia |
| S01 Lab3 | Thu 9:30am-10:20am | PENG Xinghao | PENG Xinghao |
| S01 Lab4 | Wed 5:30pm-6:20pm | WU Zekai | CHU Jinzhao |
| S01 Lab5 | Tue 5:30pm-6:20pm | PENG Xinghao | ZHANG Hengjia |
| S01 Lab6 | Wed 3:30pm-4:20pm | WU Zekai | WU Zekai |

# Labs

Session: S02 & S03

Venue: PQ603

Format: with TA face-to-face

| Lab ID | Lab Date and Hour | Week 2-7 | Week 8-13 |
|---|---|---|---|
| S02 Lab1 | Tue 3:30pm-4:20pm | ZHANG Yibin | ZHANG Yibin |
| S02 Lab2 | Thu 3:30pm-4:20pm | ZHANG Yibin | QIN Ziheng |
| S02 Lab3 | Tue 10:30am-11:20am | QIN Ziheng | QIN Ziheng |
| S02 Lab4 | Thu 10:30am-11:20am | LI Bairui | HU Xu |
| S03 Lab2 | Tue 9:30am-10:20am | SONG Haoyang | SONG Haoyang |
| S03 Lab3 | Thu 8:30am-9:20am | LI Bairui | HU Xu |
| S03 Lab4 | Thu 11:30am-12:20pm | LI Bairui | HU Xu |

# Labs

During face-to-face teaching, labs are all held in PQ603 (we need embedded boards inside that room).

We only have limited embedded boards, so several students may need to share one board.

TA should tutor the assigned lab by answering questions raised by the students.

# Course Materials

Textbooks:

**A. V. Aho, M. Lam, R. Sethi, and J.D. Ullman, "Compilers: Principles, Techniques, and Tools," Prentice Hall, 2nd Edition, 2006.**

Key A Robbings, and Steven Robbings, "Unix Systems Programming," Prentice Hall PTR, 2003.

G. Pajari, "Writing Unix Device Drivers," Addison-Wesley Publishing Company, 1993.

Supplementary Readings:

**Brian W. Kernighan, Dennis M. Ritchie, C Programming Language (2nd Ed.), Pearson, March 22, 1988.**

Textbooks in compilers and Unix programming (many, see syllabus for a list of references)

Other references (device driver, Lex, Yacc) posted on Learn@PolyU

# About the Syllabus: What is System Programming?

Types of Programs

Application programs (+ application layer libraries)

Programs for direct interaction with users

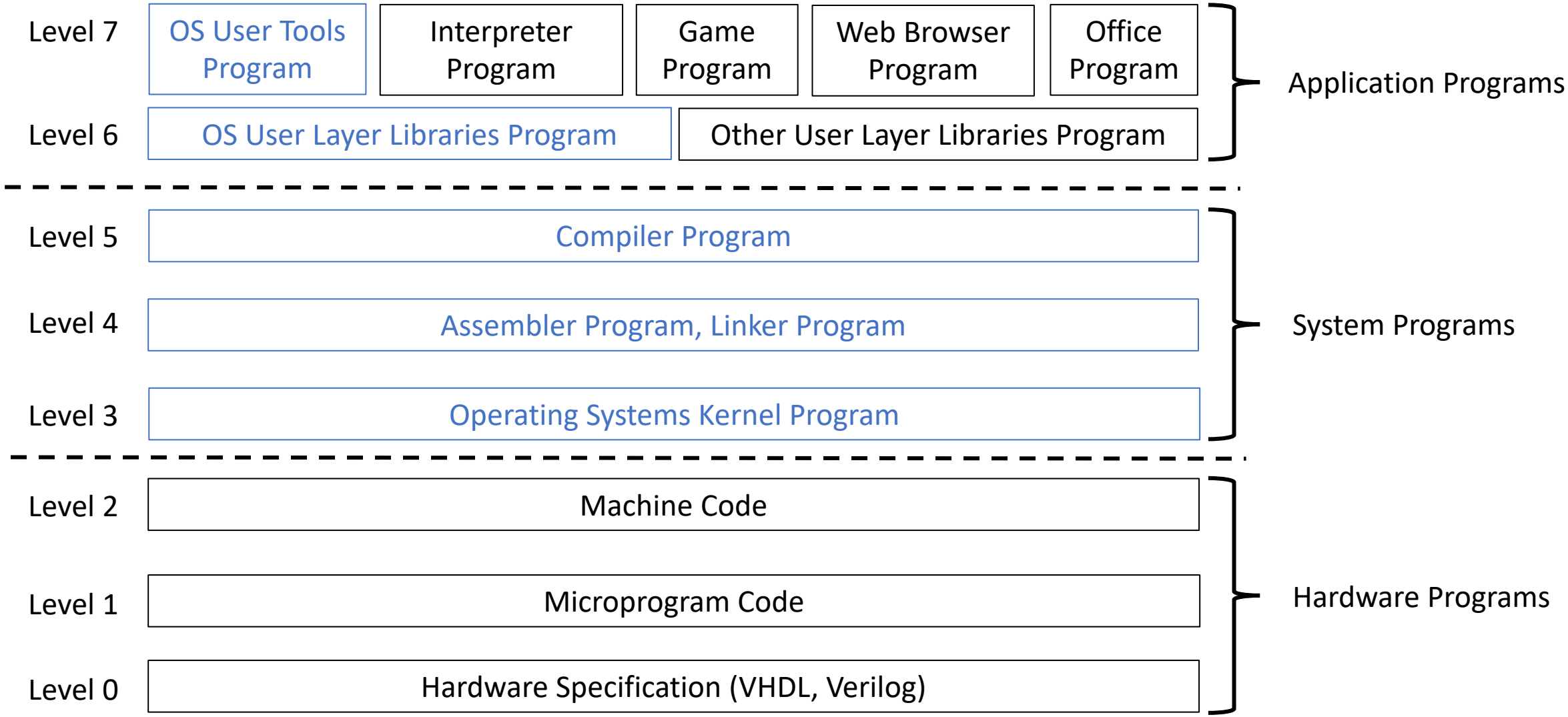
Compiler programs, assembler programs, linker programs

Tools that convert application programs to executables

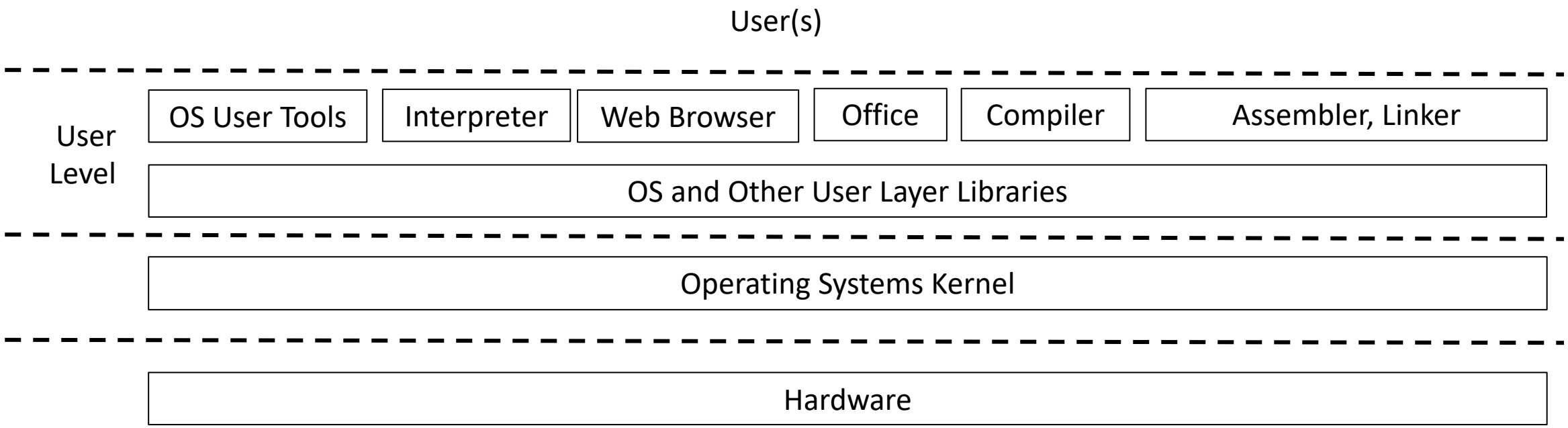
Operating systems kernel programs

Programs that interface other software with the hardware

# About the Syllabus: Layers of Programs (Development Time)



# About the Syllabus: Layers of Runtime



# About the Syllabus: Objectives of This Subject

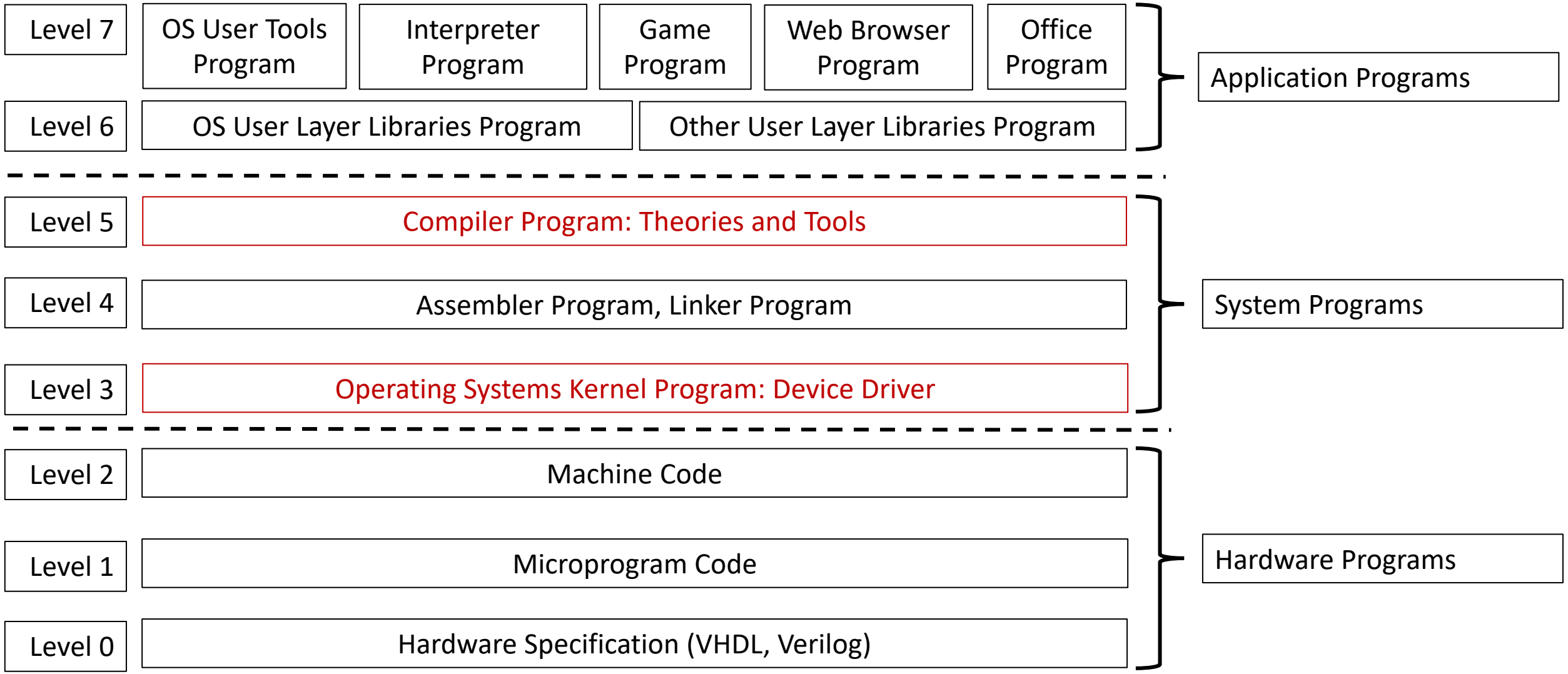
This subject introduces

- Concepts and principles of system programming;
- Methods, techniques, and tools for designing and implementing system level programs;
- Software development using OS services, and other tools.

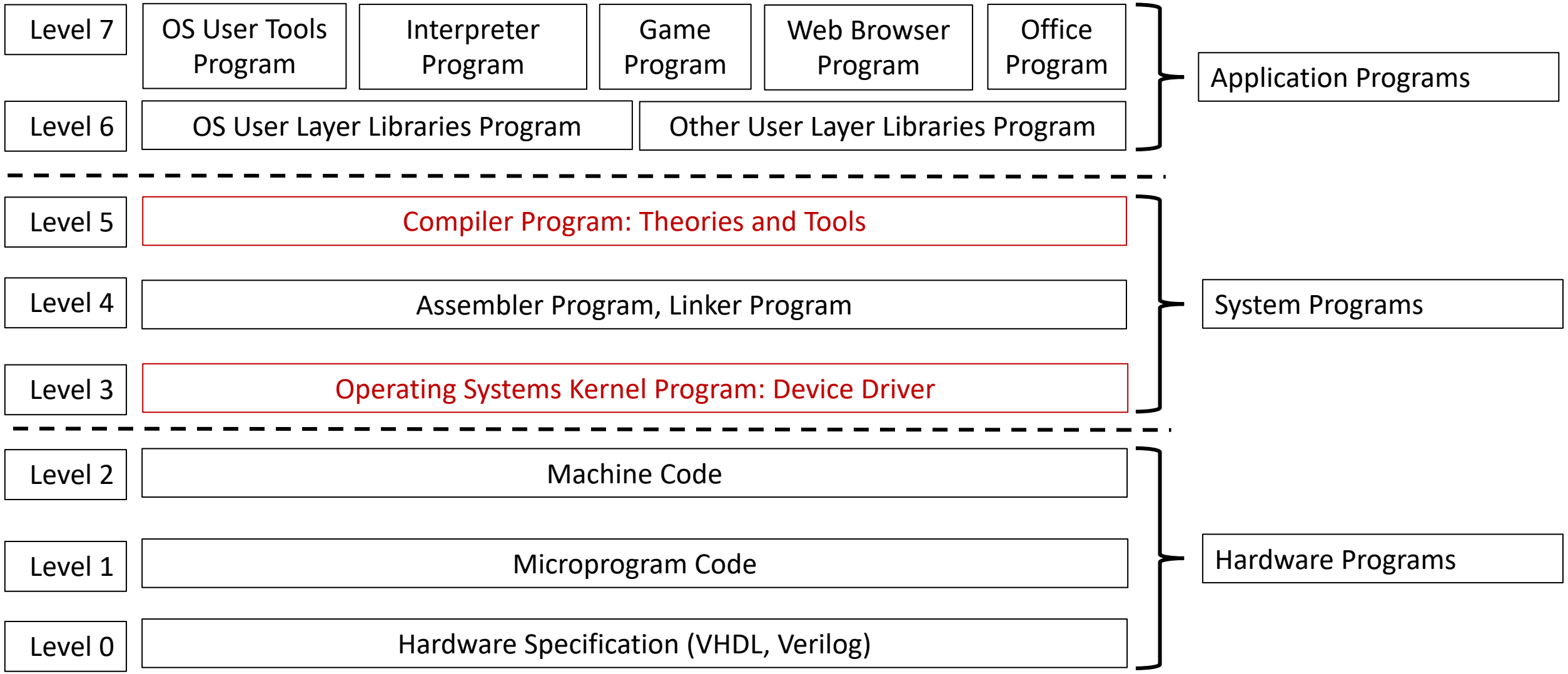
We focus on

- Operating systems kernel programs: device driver programming;
- Compiler programs: theoretical foundations, and tools.

# About the Syllabus: Objectives of This Subject



# About the Syllabus: Why learning programming (including system programming) is still important in the era of AI/LLM?



# About the Syllabus: Topics

Introduction to system software

Part I: Unix system programming – Unix device driver development

- Basic knowledge:
  - Unix programming basics, process, file systems
- Unix device driver development:
  - Basics, character device, block device
- Experimental platform:
  - Embedded Linux development environment (ARM 9)
  - Lab: LED etc.

# About the Syllabus: Topics

Part II: Compiler

- Syntax and semantics of programming languages
- Lexical analysis
- Syntax analysis
- Code generation and optimization
- Compiler tools

# About the Syllabus: Background Needed

Data structures

Programming language syntax and semantics

C programming language

Basic understandings of OS (process, file systems, etc.)

# Course Organization

